

## Code-Layout, Readability and Reusability

|                      |                             |
|----------------------|-----------------------------|
| <b>Date</b>          | 28 June 2026                |
| <b>Team ID</b>       | XXXXXX                      |
| <b>Project Name</b>  | Credit Card Fraud Detection |
| <b>Maximum Marks</b> | 5 Marks                     |

### Code-Layout, Readability and Reusability:

Dataset: Kaggle Credit Card Fraud Detection. Two inter-related CSV files are merged during preprocessing. One contains applicant demographic and financial information, while the other stores credit payment status records.

### Code Layout Checklist

| S.No | Code Quality Parameter    | Description          | Followed | Remarks             |
|------|---------------------------|----------------------|----------|---------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs | Yes      | Well formatted      |
| 2    | Proper File Structure     | Logical organization | Yes      | Clear folders       |
| 3    | Meaningful Variable Names | Purpose reflected    | Yes      | Readable            |
| 4    | Function Names            | Descriptive          | Yes      | Good naming         |
| 5    | Code Comments             | Logic explained      | Partial  | Needs more comments |
| 6    | Modular Design            | Reusable modules     | Yes      | Functions separated |
| 7    | No Redundant Code         | No duplicate code    | Yes      | Optimized           |
| 8    | Error Handling            | Graceful exceptions  | Partial  | Can improve         |

### Reusable Components / Modules

| S.No | Component / Module  | Language | Where Reused  | Reusability |
|------|---------------------|----------|---------------|-------------|
| 1    | Data Loader         | Python   | Preprocessing | High        |
| 2    | Data Cleaning       | Python   | Pipeline      | High        |
| 3    | Feature Engineering | Python   | Training      | High        |
| 4    | Model Training      | Python   | Prediction    | Medium      |
| 5    | Model Evaluation    | Python   | Testing       | Medium      |

# Overall Code Quality Assessment

| Aspect                   | Rating (1-5) | Comments           |
|--------------------------|--------------|--------------------|
| Code Layout & Structure  | 5            | Well organized     |
| Readability              | 5            | Easy to understand |
| Reusability              | 4            | Modules reusable   |
| Documentation / Comments | 4            | Adequate comments  |